

19-200-0105B

Basic Mechanical Engineering

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Syllabus

19-200-0105B/19-200-0205A BASIC MECHANICAL ENGINEERING

Course Outcomes:

On completion of this course the student will be able to:

1. Understand basics of thermodynamics and working of steam turbines
2. Understand basics of internal combustion engines, refrigeration and air conditioning
3. Gain knowledge on the working of hydraulic turbines and centrifugal pumps
4. Identify manufacturing methods encountered in engineering practice and understand mechanism of power transmission

Module I

Thermodynamics: Thermodynamics systems – open, closed and isolated systems, equilibrium state of a system, property and state, process, cycle, Zeroth law of thermodynamics- concept of temperature, temperature scales. First law – internal energy, enthalpy, work and heat, Different processes (isobaric, isochoric, isothermal, adiabatic and polytropic processes). Second law – Kelvin-planck and Claussius statements and their equivalence, Carnot Cycle (Elementary problems only). Thermodynamic properties of Steam, Steam Generator. Different types of boilers, boiler mountings and accessories. Formation of steam at constant pressure, working of steam turbines, compounding of turbines.

Module II

Internal Combustion Engines: Air standard cycles – Otto and Diesel cycles, working of two stroke and four stroke Petrol and Diesel engines, Carburetted and MPFI engines, fuel pump, fuel injector, ignition system, cooling system, lubricating system.

Refrigeration & Air-conditioning: Introduction to refrigeration and air -conditioning, Rating of refrigeration machines, Coefficient of performance, Simple refrigeration vapour compression cycle (Elementary problems only), Summer and winter air conditioning.

Module III

Hydraulic Turbines & Pumps: Introduction, Classification, Construction details and working of Pelton Wheel, Francis and Kaplan turbines, Specific speed (Definition and significance only),



Classification of water pumps, working of centrifugal pumps and reciprocating pumps (Theory of working principles only)

Power plants: Hydro-electric power plants, Thermal power plants, Nuclear power plants, Diesel power plants, Wind mills, solar energy (Working principles using schematic representations only)

Module IV

Introduction to Manufacturing Systems: Welding- different types of welding, resistance welding, arc welding, gas welding, Brazing and soldering, Different welding defects. Casting- different casting processes, sand casting, casting defects, Rolling- hot rolling and cold rolling, two high, three high, cluster rolling mills, wire drawing, forging, extrusion, Heat treatment of steel, elementary ideas of annealing, hardening, normalizing, surface hardening.

Power Transmission Methods and Devices: Introduction to Power transmission, Belt, Rope, Chain and Gear drive. Length of belt open and crossed. Ratio of belt tensions (Elementary problems only). Different types of gears (Elementary ideas only). Types and functioning of clutches.

References

1. Nag, P.K. Engineering thermodynamics. (Fifth Edition). McGraw Hill Education (India) Pvt. Ltd, New Delhi.(2013).
2. Gill, J.H. Smith Jr. and Ziurys, E.J. Fundamentals of internal combustion engines, Oxford & IBH, New Delhi.(1959)
3. Stoecker, W. F. Refrigeration and air conditioning. Tata McGraw Hill, New Delhi. (1980).
4. JagadishLal. Hydraulic machines. Metropolitan Book co, New Delhi.(1994)
5. Raghavan, V. Material science and engineering, Prentice Hall of India, New Delhi. (2004)
6. Rajendar Singh.Introduction to basic manufacturing processes and workshop technology, New Age International, New Delhi. (2006).

Module-3b: **Power Plants**





Power Plants

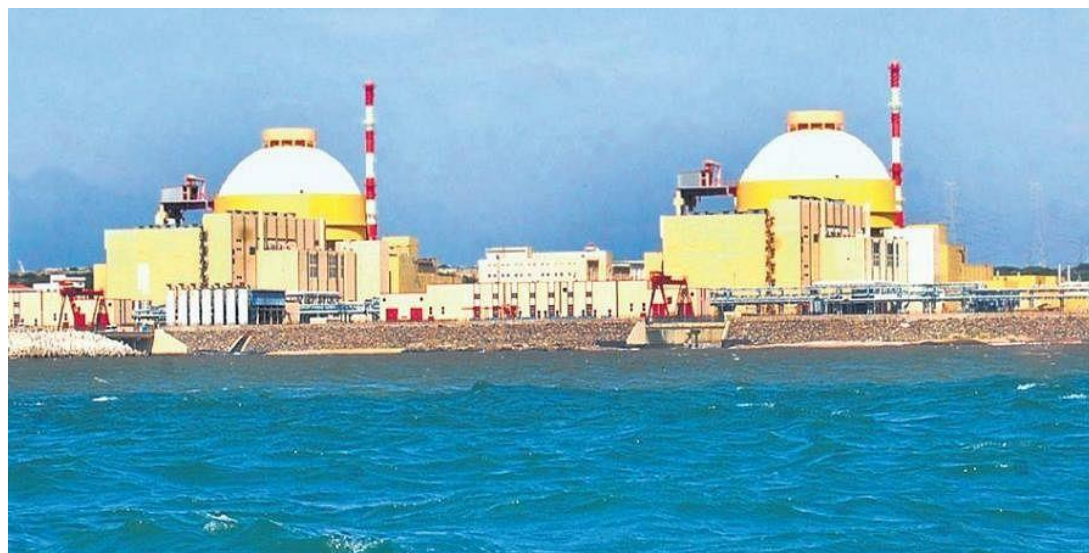


- A power plant is an industrial facility used to generate electric power with the help of one or more generators which converts different energy sources into electric power.
- Electricity is a secondary energy source - obtained from the conversion of other primary sources of energy :-
 coal, natural gas, nuclear, solar,
 hydraulic or wind energy.
- The power plant is the location in which the energy conversions take place.

Conventional Power Plants



- Fossil fuel power plants: Generates electric power by burning fossil fuels like coal, natural gas or diesel.
- Nuclear power plants: Controlled nuclear reaction is maintained to generate electricity.
- Hydroelectric power plants: Electricity is produced by building dams on suitable rivers.



Non-Conventional Power Plants

- Wind power plants: The kinetic energy of wind is used to create power.
- Solar power plants: Generates power by collecting solar radiation.
- Geothermal power plants: Uses the natural heat found in the deep levels of the earth to generate electricity.
- Biomass power plants: Natural organic matter is burnt to produce electricity.



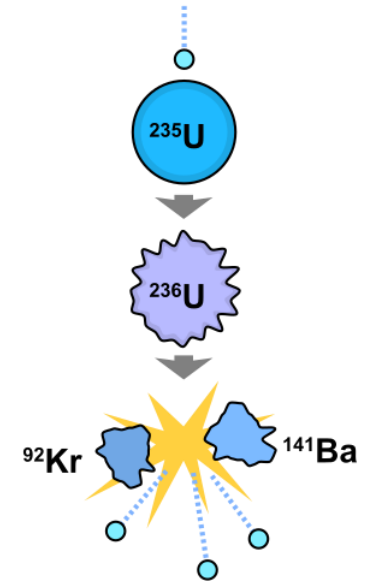
Power Plants



- Hydro-electric power plants
- Thermal power plants
- Nuclear power plants
- Diesel power plants
- Wind power plants
- Solar power plants

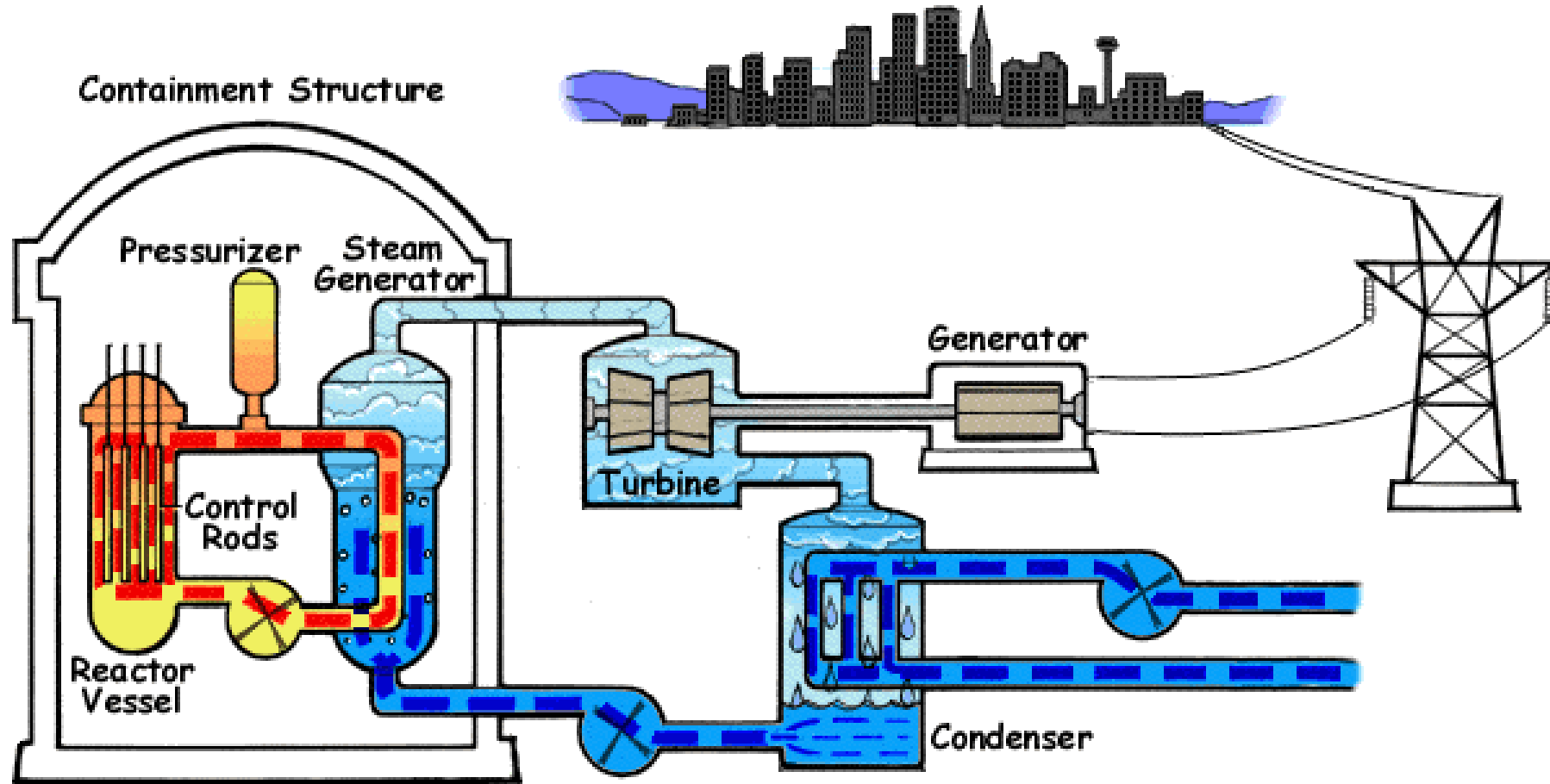
Nuclear Power Plant

- Nuclear power is the fifth-largest source of electricity (1.8%) in India after coal, gas, hydroelectricity and wind power¹
- Total installed capacity of 6,780 MW, 7 more reactors are under construction with a combined generation capacity of 4,300 MW¹
- Based on Nuclear fission
- Main components are :
 - Reactor core:- Produces heat due to nuclear fission
 - Control rod:- To control fission rate
 - Coolant:- Removes energy from core and raise steam
 - Turbine:- Thermal energy to mechanical energy
 - Electric generators:- Mechanical energy to electrical energy



Schematic of nuclear fission reaction

Working of Nuclear Power Plant



Nuclear Power Plants



Advantages

- Produces no polluting gases.
- Does not contribute to global warming.
- Low fuel quantity reduces mining and transportation effects on environment.
- Power station has very long lifetime.

Nuclear Power Plants



Disadvantages

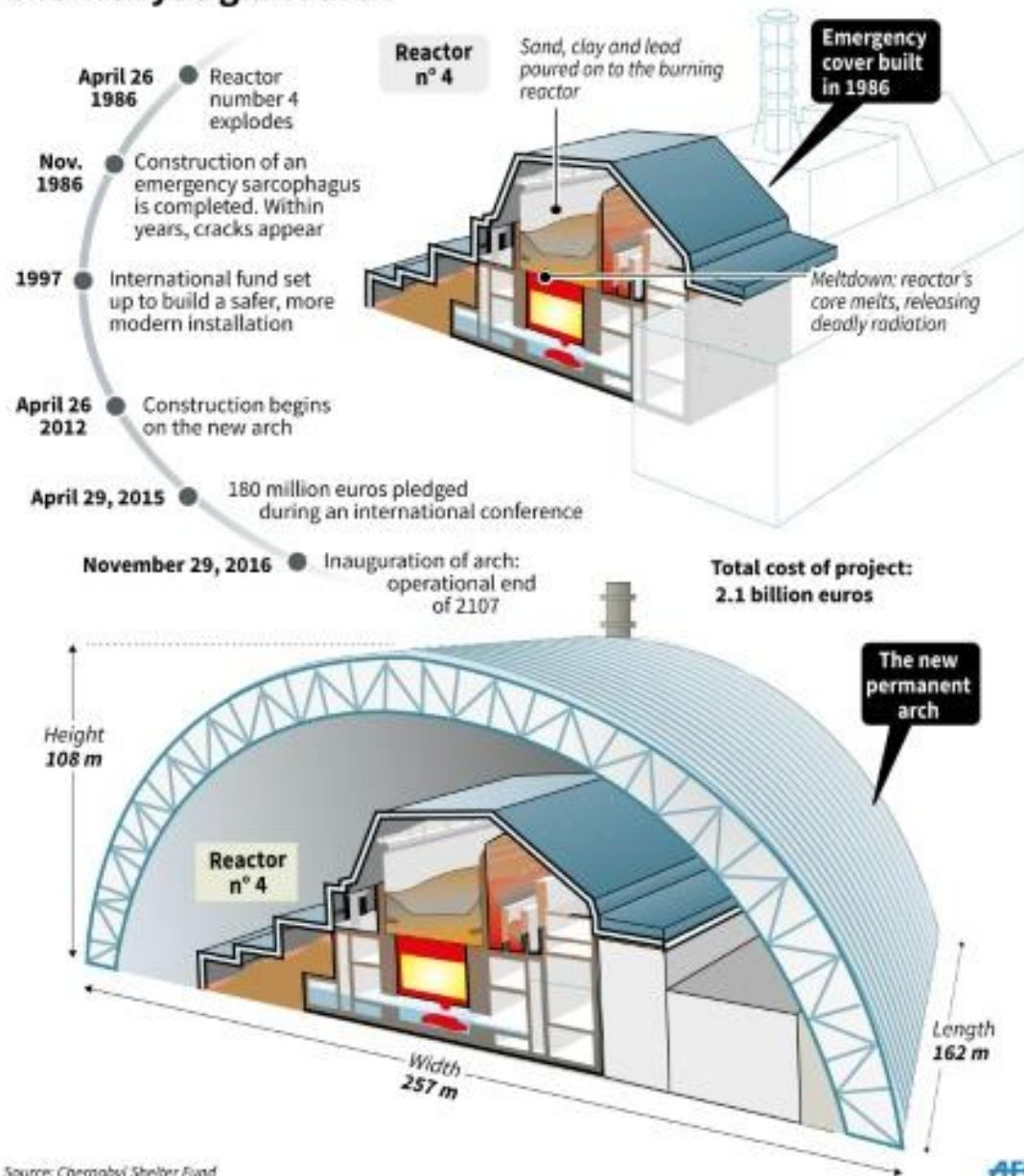
- Waste is radioactive and safe disposal is very difficult and expensive.
- Costs of building and safely decommissioning are very high.
- Cannot react quickly to changes in electricity demand.
- Local thermal pollution from wastewater affects marine life.

Accidents

- Chernobyl 1989
- Fukushima 2011



Chernobyl's giant arch

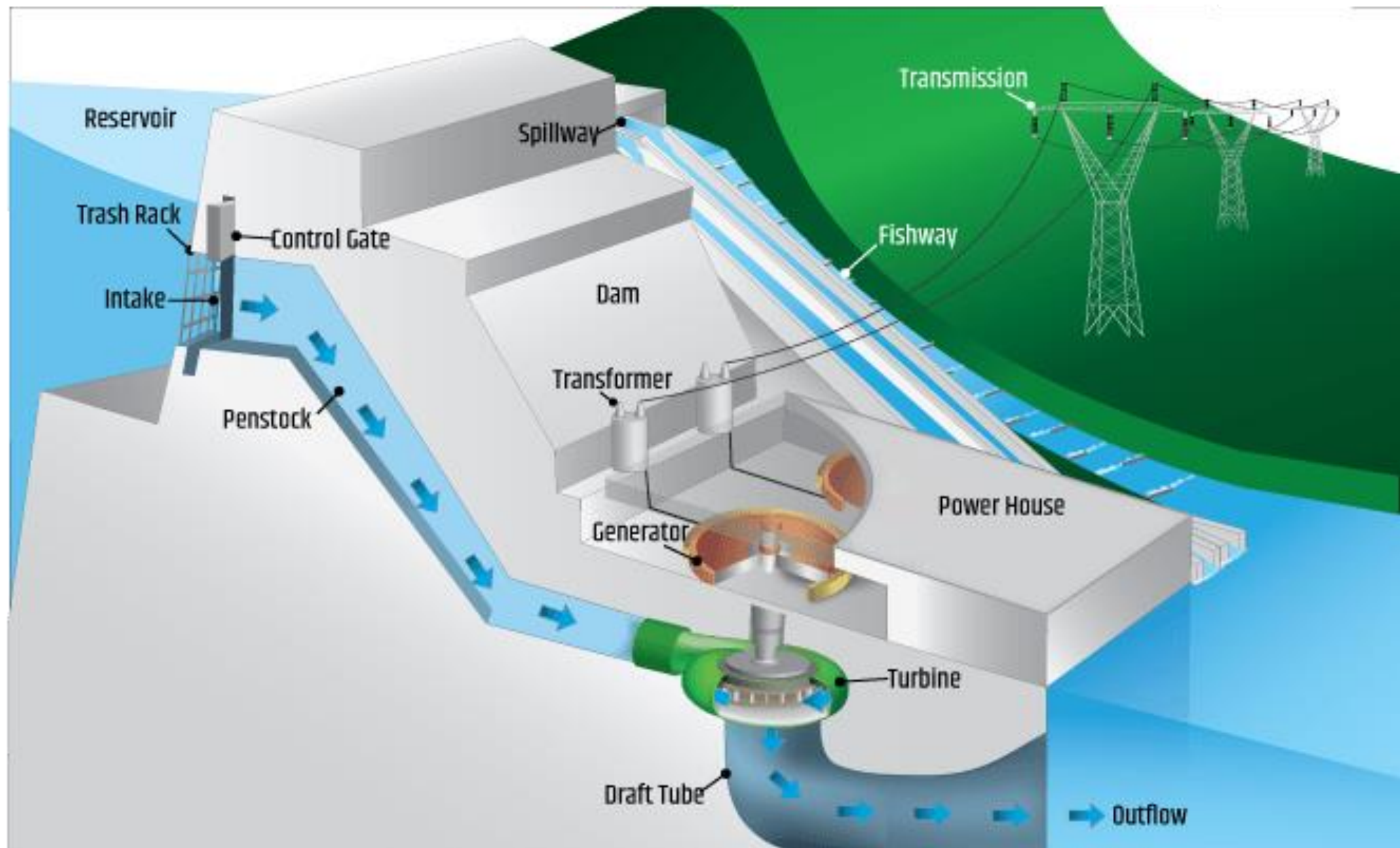


Hydroelectric Power Plant

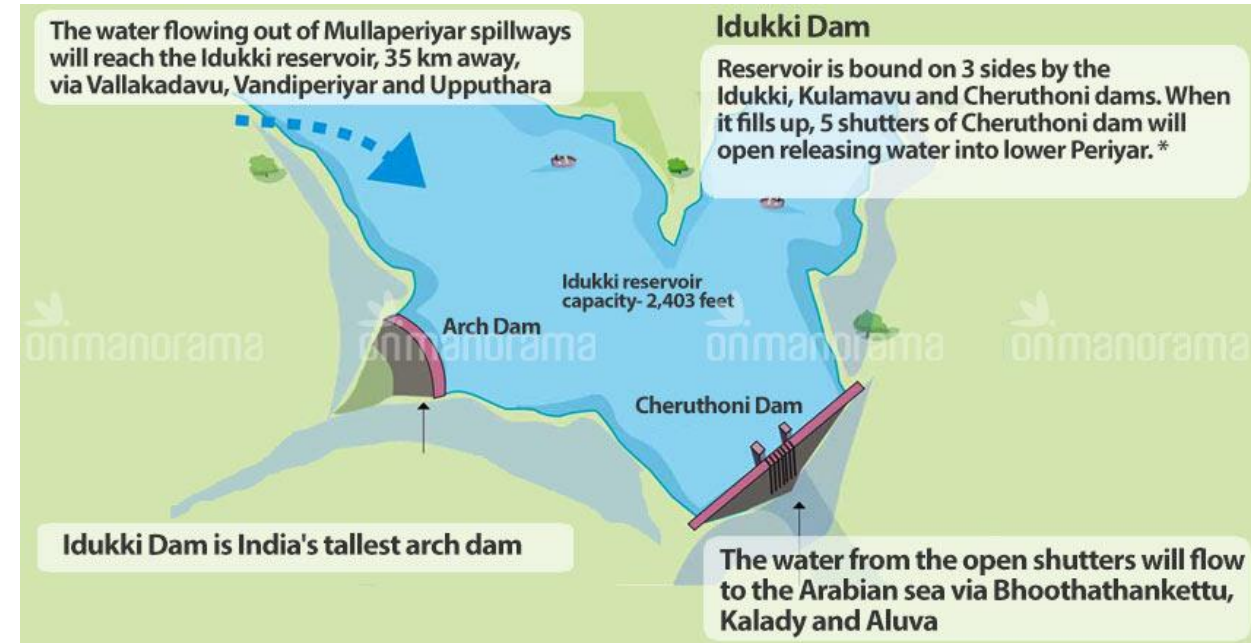


Hydro - Electric Power Plants

- Hydro energy:- Energy generated by the force of water.
- Production of electricity depends upon:-
 - Water level from the ground
 - Flow rate of water
 - Efficiency of turbine



Idukki Dam



Hydro - Electric Power Plants

Advantages

- The most important advantage is that it keeps the environment clean
- Hydro energy can be stored in high level reservoirs and used whenever necessary.
- Electricity is produced at a constant rate.
- The life of dams and power plants last for many decades.
- The water can be used for supplying
 - Drinking water
 - Irrigation
 - Water sports
 - Industries
 - Power plants

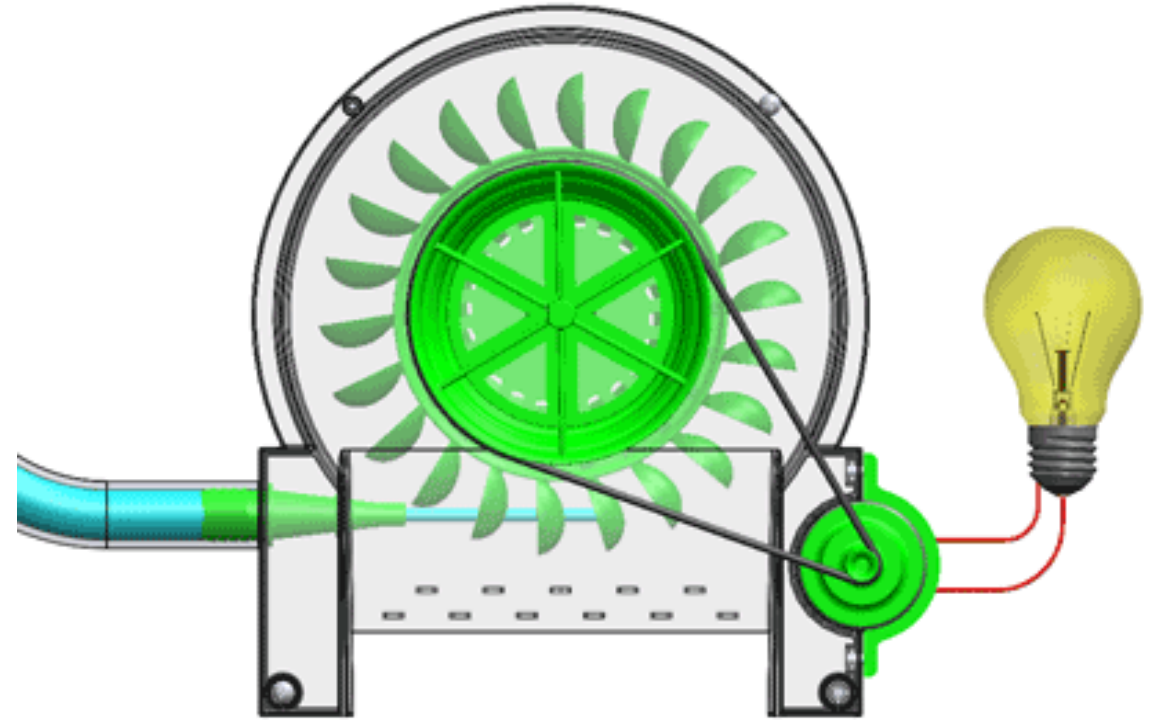
Hydro - Electric Power Plants

Disadvantages

- Constructing hydroelectric power plants and dams are very expensive.
- Requires large area.
- Planning a hydro electric power plant and reservoir may lead to the disruption of the lives of people as they have to relocate to another area.

Idukki Dam Details

Commission date	1975
Turbines	6x130 MW Pelton Turbines
Installed Capacity	780 MW



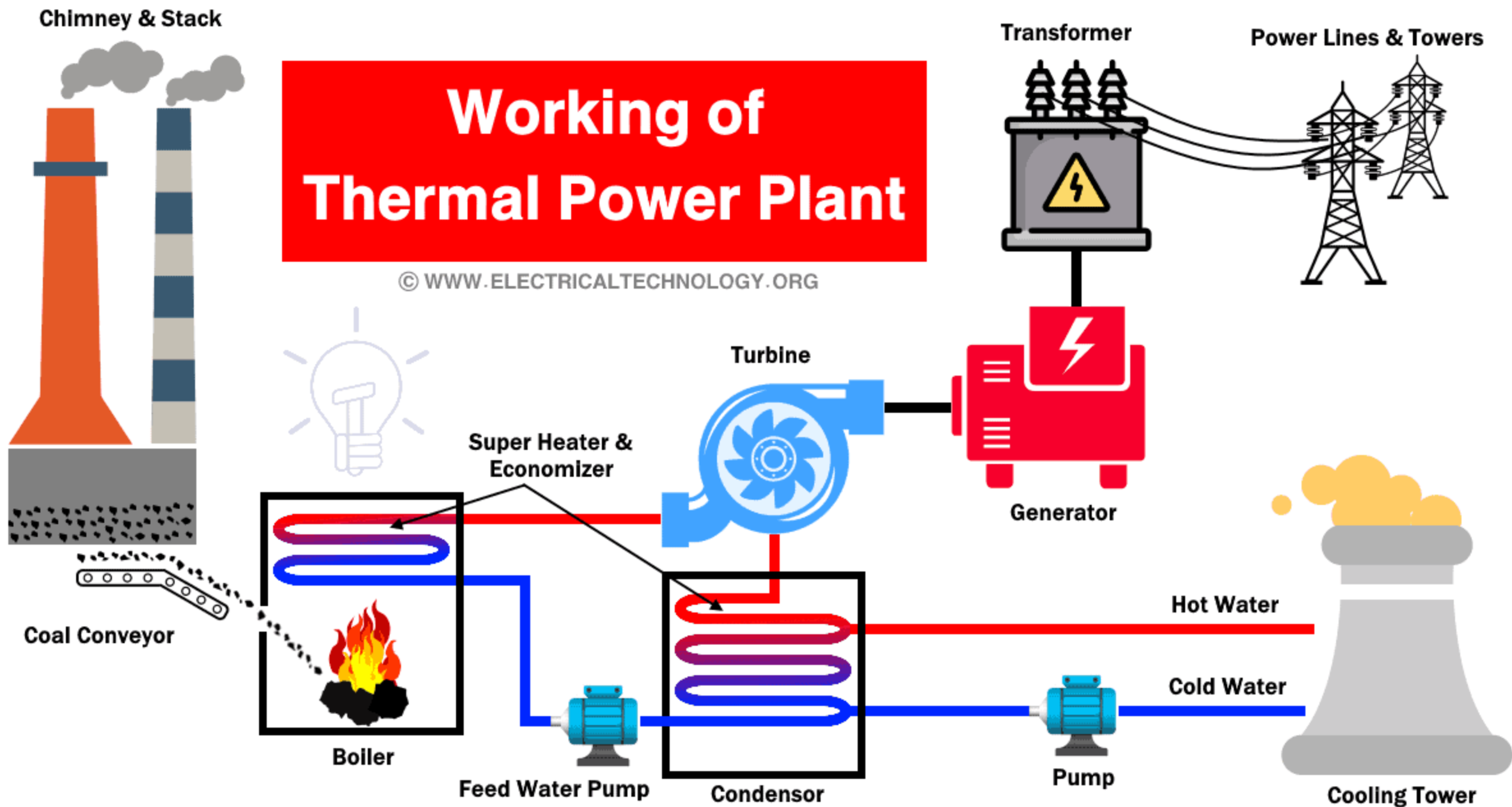
Thermal Power Plant



Thermal Power Plant



- ❑ A generating station which converts heat energy of coal combustion into electrical energy is known as a steam power station
- ❑ The heat produced for burning of coal & with the help of water steam is produced. This produced steam flow towards turbine i.e. kinetic energy of the steam which converted into mechanical energy of the turbine. The input steam drives the prime mover or turbine, simultaneously the generator also start to rotate. At that time mechanical energy is converted into electrical energy.
- ❑ Thermodynamic cycle:- Rankine cycle



Thermal Power Plants



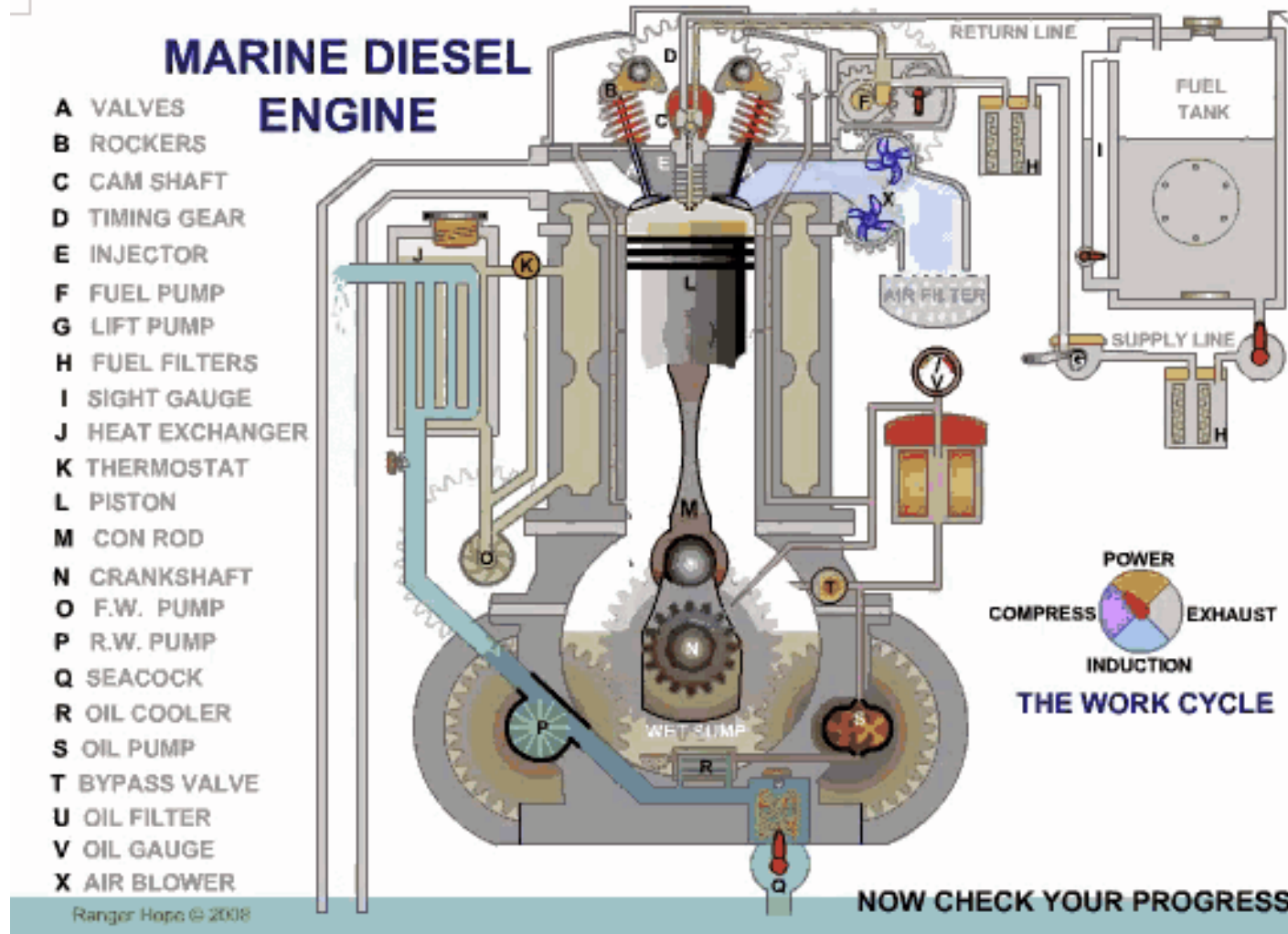
- Advantages:
 - Less initial cost as compared to other generating stations.
 - It requires less land as compared to hydro power plant.
 - The fuel (i.e. coal) is cheaper.
 - The cost of generation is lesser than that of diesel power plants.

Thermal Power Plant



- Disadvantages:
 - It pollutes the atmosphere due to the production of large amount of smoke. This is one of the causes of global warming.
 - The overall efficiency of a thermal power station is low (less than 30%).

Diesel Power Plant

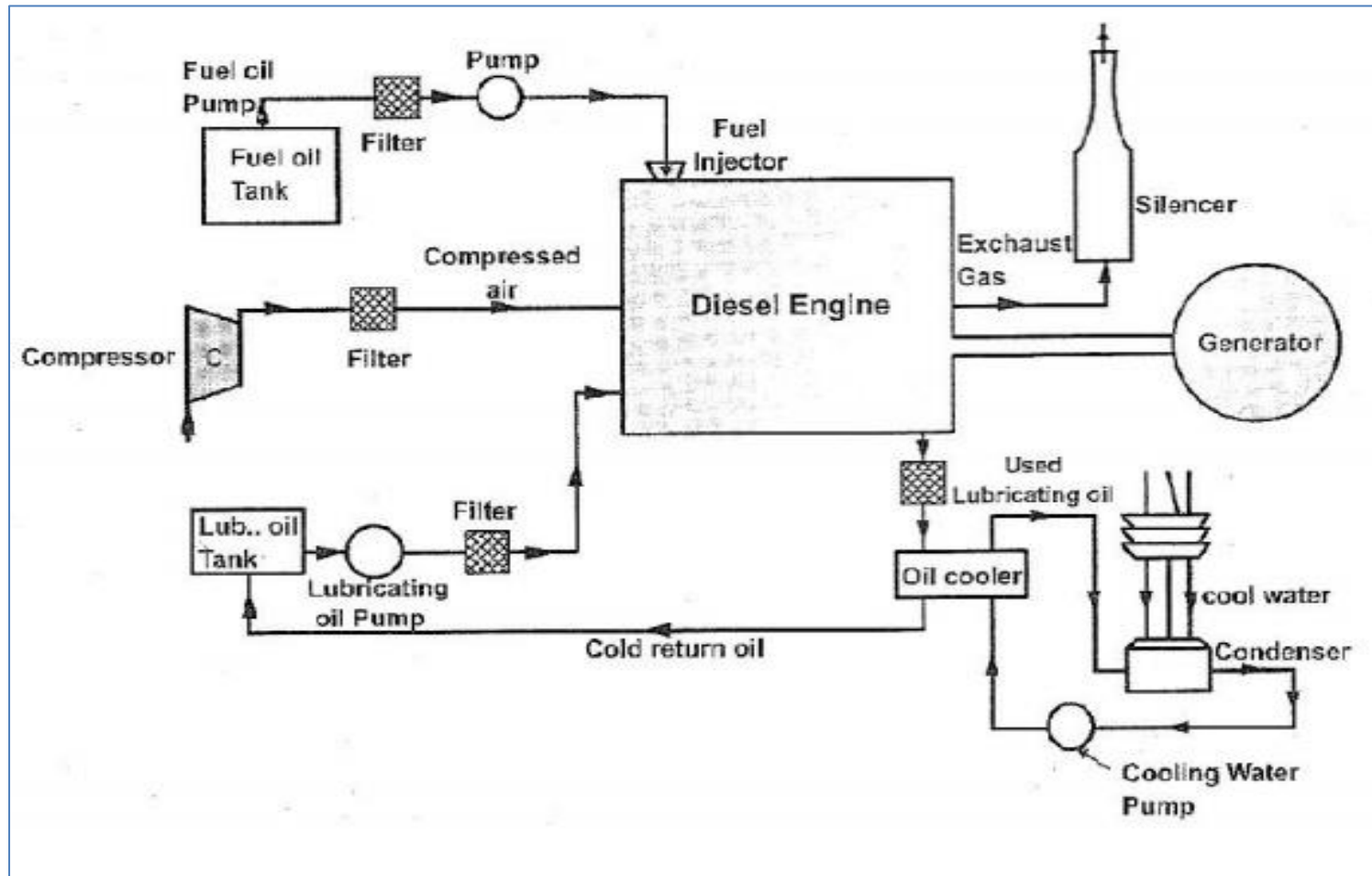


Diesel Power Plant



- A Diesel power station (also known as Stand-by power station) uses a diesel engine as prime mover for the generation of electrical energy. This kind of power station can be used to produce limited amounts of electrical energy.
- The diesel burns inside the engine and the products of this combustion act as the working fluid to produce mechanical energy. The diesel engine drives alternator which converts mechanical energy into electrical energy

Diesel Power Plant



Diesel Power Plant



- Advantages
 - Quick installation and commission
 - Quick starting and easy pick-up of loads
 - Easy operation
 - Required minimum labours
 - Simple fuel handling
 - Efficient ash disposal
 - High level of efficiency

Diesel Power Plant



- Disadvantages
 - Limited power generation capacity.
 - Expensive due to the use of diesel.
 - High maintenance cost.
 - Relatively short life.

Example: Nallalam Diesel Power Plant

Project Details:

- Project Name – Nallalam Diesel Power Plant
- Commission Date – 1999-09-01
- Total Installed Capacity in MW – 128 MW
- Status of Plant - Operational
- Power Plant Used for – Peaking Load
- Type of Plant – Oil Engine
- Type of Fuel – LSHS Oil
- Boiler Configuration –
- Name of Operator - Kerala State Electricity Board (KSEB)
- Name of Substation –
- Diesel Engine / Boiler Manufacturer- Wartsalia
- Diesel Engine / Boiler Model/Type - Wartsalia 46 NSD
- Address – Kozhikode, Kerala, India
- District – Kozhikode
- State – Kerala
- Contact Number - +91 (471) 2446480
- Email –
- Website – kseb.in
- Beneficiary States – Kerala and its neighbouring states
- Unit Sizes – 458.12 MW
- Units Operational – 8
- Unit Commissioned – All units are commissioned

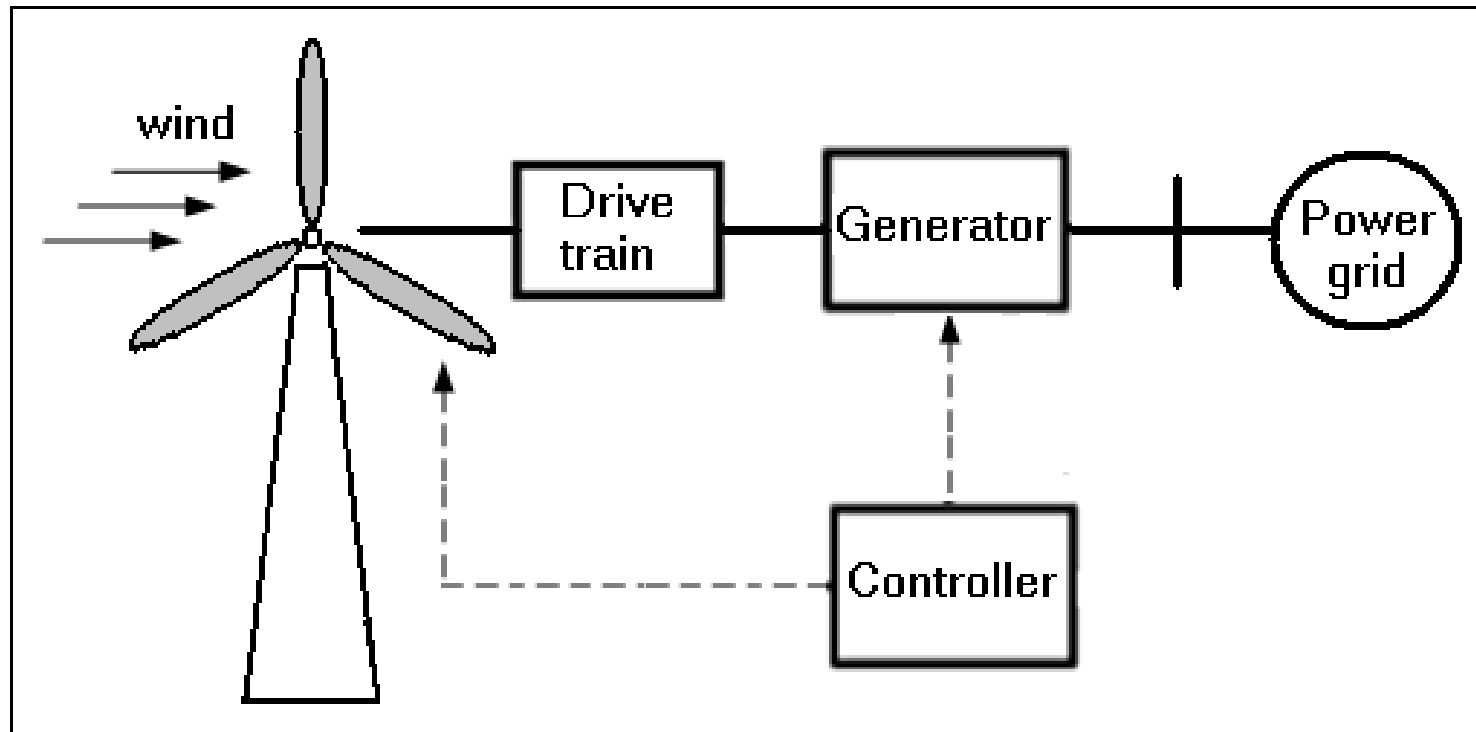


Wind Power Plant



Wind Power Plant

- Wind turbines operate on a simple principle. The energy in the wind turns two or three propeller-like blades around a rotor. The rotor is connected to the main shaft, which spins a generator to create electricity



Wind Power Plant



- Advantages
 - Clean & Environment friendly Fuel source
 - Renewable & Sustainable
 - Cost Effective
 - Industrial and Domestic Installation
- Disadvantages
 - Fluctuation of Wind
 - Noise and aesthetic pollution
 - Not a profitable use of land

Solar Power Plant



Solar Power Plant



- A solar power plant is any type of facility that converts sunlight either directly, like Photovoltaics, or indirectly, like Solar Thermal plants, into electricity.
- When photons from sunlight hit the semiconductor material free electrons are generated which can then flow through the material to produce a direct electrical current - known as the photo - effect in physics.
- DC to AC - convertors
- Can use the photo-effect directly without the need of other processes or devices.

Solar Thermal Power Plants

- These kinds of solar thermal power plant work by focusing sunlight from long parabolic mirrors onto receiver tubes that run the length of the mirror at their focal point. This concentrated solar energy heats up a fluid that continuously flows through the tubes.
- This heated fluid is then sent to a heat exchanger to boil water in a conventional steam-turbine generator to generate electricity.



Solar Thermal Power Plant in Middle East

Solar Power Plant



Advantages

- Renewable Energy Source
- Reduces Electricity cost
- Diverse Applications
- Low Maintenance Costs
- Technology Development

Disadvantages

- Weather Dependent
- Solar Energy Storage is Expensive
- Uses a Lot of Space

Thank You